Week 11 9–13 November 2026

Subject: Computer science Unit: A2.1 Network fundamentals Grade: 11DP SL

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents المصادر و الروابط المخصصة لأولياء الأمور
Lesson 1 Networks and infrastructure 60 minutes	A2.1.1–A2.1.2 Describe purposes and characteristics of LAN, WAN, PAN and VPN. Describe the purpose, benefits and limitations of the internet, cloud computing, distributed systems, edge computing and mobile networks.	5 min: retrieve OS networking. 15 min: classify four network types. 20 min: expert pairs compare five infrastructures using the notes. 15 min: apply to WWW, blockchain, traffic lights and a school. 5 min: exit check. Support: comparison grid and glossary. Extend: justify cloud/edge trade-offs in a school scenario.	Notes A2.1.1 and A2.1.2; IB guide p. 31; network/infrastructure cards. Preparation: skim both notes and highlight unfamiliar terms. Homework: complete the five-infrastructure comparison with one benefit and limitation each.
Lesson 2 Network devices 60 minutes	A2.1.3 Describe NIC, switch, WAP, router, modem, hardware firewall and gateway functions. Annotate how these devices support a school network and map their principal roles to TCP/IP layers, allowing for multifunction devices.	5 min: retrieve network types. 15 min: model seven device roles. 20 min: pairs annotate a school diagram and trace local versus internet traffic. 15 min: match device roles to TCP/IP layers and answer a scenario question. 5 min: exit check. Support: labelled icons and layer reference. Extend: distinguish router/gateway and integrated hardware.	Note A2.1.3; IB guide p. 31; school-network diagram; device/layer cards. Homework: explain switch versus router and modem versus WAP; revise all seven device functions.
Lesson 3 Network protocols 60 minutes	A2.1.4 Describe TCP and UDP transport functions, and HTTP, HTTPS and DHCP application functions. Choose suitable protocols for reliable transfer, live communication, secure web access and automatic configuration; explain how they work together.	5 min: retrieve device roles. 15 min: model protocols and TCP/UDP trade-offs. 20 min: sort scenarios; trace DHCP configuration followed by secure web access. 15 min: individual Paper 1-style responses and peer review. 5 min: exit check. Support: protocol-purpose/layer grid and sentence stems. Extend: explain why HTTPS does not guarantee trustworthy content.	Note A2.1.4; IB guide p. 31; protocol cards; request/response and DHCP diagrams. Homework: complete a short five-protocol revision table and justify TCP/UDP choices in two scenarios.
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	Prepare comparison grids before teaching. Review exit checks and revisit misconceptions: VPN is not a geographical-size category; internet ≠ WWW/cloud; switch uses MAC, router uses IP. UDP does not guarantee delivery/order; HTTPS protects transport, not content trustworthiness. A2.1.3 requires device-to-layer mapping at SL; detailed A2.1.5 model teaching is HL only.		

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Technology Integration دمج التكنولوجيا	Use a digital network diagram and device/layer cards to visualise traffic paths. Students complete shared comparison grids and protocol scenarios; paper versions remain available.
Cross-curricular links الروابط بين المواد الدراسية	Geography: coverage and connectivity. Physics: signals and wireless transmission. Business: infrastructure cost, scalability and availability. Citizenship: privacy and responsible use of shared networks.
External Assessment Integration دمج التقييمات الخارجية	Paper 1 preparation: describe network characteristics, benefits/limitations, device functions and protocol purposes. Use individual short responses, technical vocabulary checks and scenario-based feedback.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	Thinking: classify systems and justify choices. Communication: explain a network path to a partner. Research: use the supplied notes to extract evidence. Self-management: complete preparation and act on exit-check feedback.
Standards المعايير	IB Computer science guide, first assessment 2027, p. 31: A2.1.1–A2.1.4. Sources: four attached notes on networks, digital infrastructures, devices and protocols.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	TOK: Does easier access to information make knowledge more reliable? Distinguish secure HTTPS transmission from the accuracy of website content; consider what further evidence would establish trust.
Success Criteria معايير النجاح (Measurable)	MUST: Distinguish LAN/WAN/PAN/VPN; give one benefit and limitation for each of five infrastructures; identify seven devices and five protocols. SHOULD: Describe device roles and principal TCP/IP layer mappings; annotate a school network; justify two protocol choices using technical features. COULD: Recommend an infrastructure/device/protocol combination with a reasoned trade-off and explain how multiple components cooperate.
National Identity Domain مجالات الربط بالهوية الوطنية	Context: design a hypothetical UAE school network connecting classrooms, shared resources and remote services; compare local and inter-campus connections.
	Values: discuss privacy, secure communication, equitable access and responsible sharing of school network resources.
	Application: compare cloud and edge processing for hypothetical smart traffic lights or building sensors in a UAE setting; explain latency and connectivity trade-offs.